SIDEBAR **Justifying binary search with information theory**

I remember when I first learned about the algorithm, the strategy of searching in the middle of an array seemed, although reasonable, quite unique and "out of nowhere." However, we have just learned to derive the same solution from an information-theoretic perspective, which concretely quantifies the idea of ruling out half of our search space in the service of gaining as much information as possible, or reducing as much entropy as possible.

The application of information theory and entropy doesn't stop at binary search. As we saw, the procedure we went through is generalizable to other decision-making problems: if we could model the problem of interest in terms of probability distributions for unknown quantities, actions we can take, and how the distributions can be updated when an action is taken, then we can once again choose the information-theoretically optimal action that reduces our uncertainty about the quantity of interest the most.

NOTE **Entropy search as a search for the truth**

You can think of this entropy-based search strategy as a search for the truth, in which we choose to perform the experiment that helps us learn the most about something we care about.

For the remainder of this chapter, we will learn to apply this idea to Bayesian optimization and implement the resulting entropy search policy with BoTorch.

6.2 Entropy search in Bayesian optimization

Taking the same approach presented in the previous section, we obtain entropy search policies in Bayesian optimization (BayesOpt). The main idea is to choose our actions, our experiments, so that we can reduce the most amount of entropy in the posterior distribution of what we care about. In this section, we will first discuss how to do this on a high level and then move on to implementation in BoTorch.

6.2.1 Searching for the optimum using information theory

In our remote control example, we aim to search for the remote within an apartment and thus want to reduce the entropy of the distribution of the location of the remove. In binary search, the process is similar: our target is the location of a specific number we'd like to search for within a list, and we want to reduce the entropy of the distribution of that number. What we can use to be our target in BayesOpt and how we can leverage information theory to aid the search process are what we need to determine in order to design an entropy search policy, which we will learn how to do here.

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